

Repechage

Agentic Payment Recovery

for Failed Razorpay Checkouts

AI reasons → **Policy decides** → **Execution is controlled**

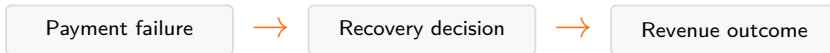
Ayushi Singh

Razorpay AI Buildathon — Track 03

The Problem



The opportunity existed. The recovery decision didn't.



Agenda

01
The Problem

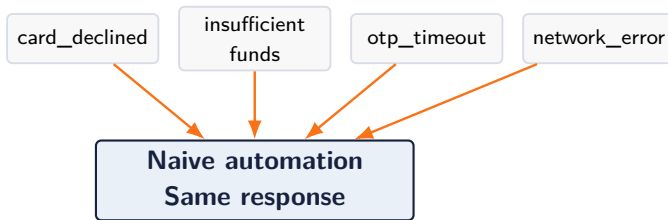
02
The Solution

03
The Architecture

04
The Evidence

- Why failed payments become lost revenue
- How Repechage makes controlled recovery decisions
- What the baseline and agent experiment revealed

Payments Fail for Different Reasons



- Wasted recovery attempts
- Missed recoverable payments
- No clear stopping point

The Opportunity is Measurable

₹98,24,112

Total transaction value

66.6%

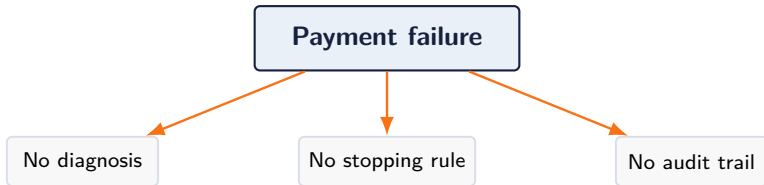
Revenue flagged at-risk

₹65,38,889

Potentially recoverable

- 1,000-event benchmark
- 438 genuinely recoverable events
- Recovery value depends on choosing the right action

Why Generic Automation Fails

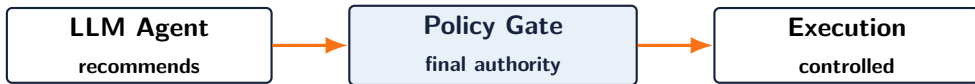


Automation can execute without understanding the recovery decision.



An agent that reasons about the recovery decision.

AI Does Not Get the Final Say



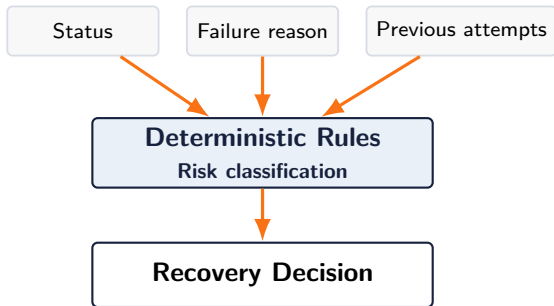
AI reasons. Policy decides. Execution is controlled.

System Architecture



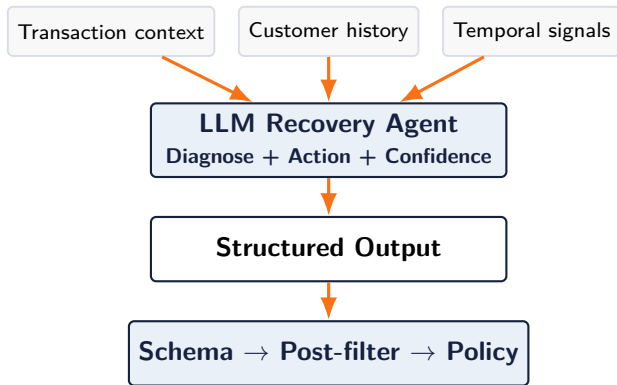
- FastAPI handles pages, authentication and webhooks
- PostgreSQL stores transaction, decision and execution history
- Decision layer separates reasoning from execution

Baseline Model — Deterministic Recovery

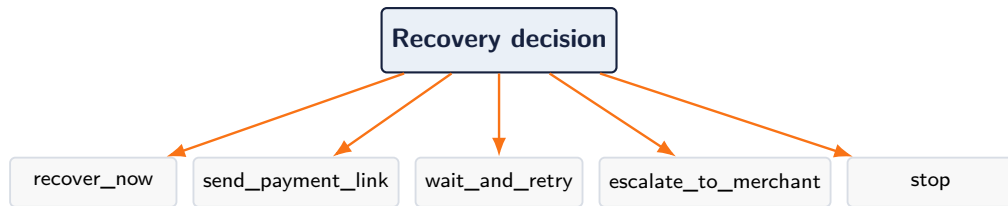


Explainable · Zero LLM cost · Same policy constraints

Agent Model — Contextual Recovery



The Agent Has a Bounded Action Space



Five actions. No free-form execution.

Safety Lives in the Policy Layer



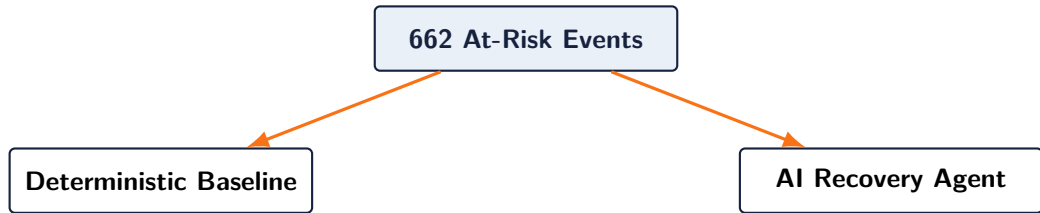
- Attempts $\geq 3 \rightarrow$ STOP
- Amount $> ₹5,000 \rightarrow$ ESCALATE
- Invalid or low-confidence output $\rightarrow$ NO EXECUTION

Every Decision Leaves a Trace



- What happened?
- Why was the action selected?
- What action was actually executed?

Same Events. Same Economics. Two Systems.



- Same 662 at-risk events
- Same simulation economics
- Ground truth hidden from both systems

The Experiment

Deterministic Baseline

662 candidates

438 recoveries

₹42,88,918 recovered

224 bad interventions

AI Recovery Agent

408 candidates

298 recoveries

₹25,76,773 recovered

110 bad interventions

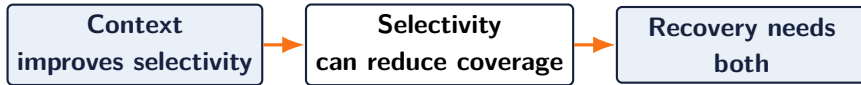
VS.

What the Experiment Showed

AI hit-rate
73.0%

Baseline hit-rate
66.2%

- AI acted on 408 candidates
- Baseline acted on 662 candidates
- The experiment exposed a precision–coverage tradeoff



Improve opportunity coverage without sacrificing safety.

From Failure to Controlled Recovery



- Bounded actions
- Policy-controlled execution
- Every decision auditable

Repechage

Agentic Recovery for Failed Payments



AI reasons. Policy decides. Execution is controlled.

Ayushi Singh